University of Engineering & Technology Peshawar

Programming Fundamentals (C++) Lab

LAB REPORT 3

STUDENT NAME	MUHAMMAD SAHAL AYUB
ETEA ID:	4042
COUSRE:	EE-170LCOMPUTER PROGRAMMING LAB
DATE:	3/2/26
LAB TITLE:	COMPARSION ,LOGICAL AND CONDITIONAL OPERATION IN C++

SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

The objective of this lab is to practice the use of:

- Comparison/Relational operators
- Logical operators
- Conditional statements: if, if-else, and else-if in C programming

Theory:

1. Comparison /Relational operator:

Comparison operators are used to compare two values in C. They return either true (non-zero) or false (0).

== : Equal to

!= : Not equal to

> : Greater than

< : Less than

>= : Greater than or equal to

<= : Less than or equal to.

Example:

```
if(a > b) {
    cout << "a is greater than b";
}
```

2. Logical Operators:

Logical operators are used to combine multiple conditions:

&& : Logical AND (true if both conditions are true)

|| : Logical OR (true if at least one condition is true)

! : Logical NOT (inverts the result)

Example:

```
if(a > 0 && a < 100) {
  cout << "Number is between 1 and 99";
}
```

3. Conditional Statements

Conditional statements control the flow of the program based on conditions.

a) if statement – executes code block if condition is true:

```
if(a > 0) {
   cout << "Positive number";
}
```

b) if-else statement – provides two paths:

```
if(age >= 18)

   cout << "Eligible to vote";

else

   cout << "Not eligible";
```

c) else-if statement – handles multiple conditions:

```
if(a > 0)
    cout << "Positive";
else if(a < 0)
    cout << "Negative";
```

```
else
    cout << "Zero";
```

Lab Tasks & Programs

Task 1: Comparison/Relational Operators

Description:

Compare two numbers and display the result.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num1, num2;
    cout << "Enter first number: ";
    cin >> num1;
    cout << "Enter second number: ";
    cin >> num2;

    if(num1 == num2)
        cout << "Both numbers are equal." << endl;
    else if(num1 > num2)
        cout << "First number is greater than second number." << endl;
    else
        cout << "First number is less than second number." << endl;
```

```
    return 0;
}
```

Output:

```
Enter first number: 10
Enter second number: 20
First number is less than second number.
```

Task 2: Logical Operators

Description: Check if a number is greater than 0 and less than 100.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;

    if(num > 0 && num < 100)
        cout << "Number is greater than 0 and less than 100." << endl;
    else
        cout << "Number does not meet the conditions." << endl;

    return 0;
}
```

Output:

```
 Enter an integer: 50

Number is greater than 0 and less than 100.
```

Task 3: If Statement:

Description: Check if a number is positive.

Code:

```
#include <iostream>

using namespace std;


int main() {

    int num;

    cout << "Enter an integer: ";

    cin >> num;


    if(num > 0)

        cout << "The number is positive." << endl;

    else

        cout << "The number is not positive." << endl;


    return 0;

}
```

Output:

```
Enter an integer: -5

The number is not positive.
```

Task 4: If-else Statement:

Description: Check voting eligibility (age ≥ 18).

Code:

```
#include <iostream>
using namespace std;

int main() {
    int age;
    cout << "Enter your age: ";
    cin >> age;

    if(age >= 18)
        cout << "You are eligible to vote." << endl;
    else
        cout << "You are not eligible to vote." << endl;

    return 0;
}
```

Output:

```
Enter your age: 20
You are eligible to vote.
```

Task 5: Else-if Statement:

Description: Check if a number is positive, negative, or zero.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;

    if(num > 0)
        cout << "The number is positive." << endl;
    else if(num < 0)
        cout << "The number is negative." << endl;
    else
        cout << "The number is zero." << endl;

    return 0;
}
```

Output:

```
Enter an integer: 0
The number is zero.
```

Conclusion:

- Learned comparison and logical operators in C.
- Practiced conditional statements (if, if-else, else-if).
- Understood how programs make decisions based on conditions.

- Gained experience in structuring programs logically to solve problems.